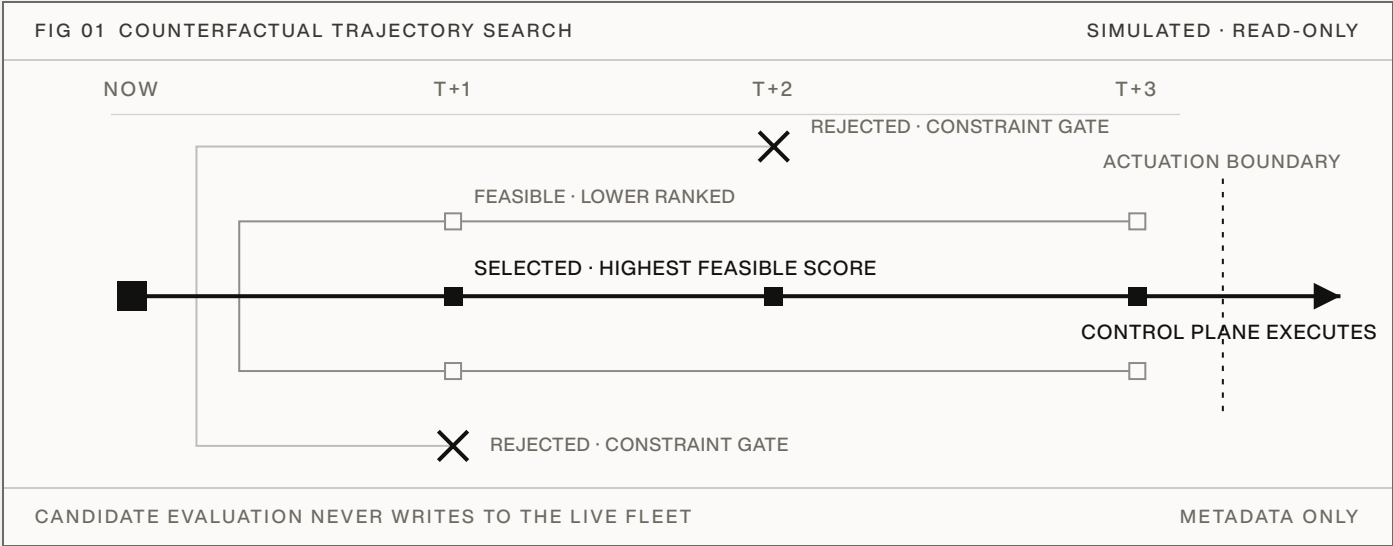


Choose the best fleet trajectory, not just the best next action.

ONE LIVE FLEET STATE · MANY COUNTERFACTUAL FUTURES · ONE SAFE TRAJECTORY SELECTED BEFORE EXECUTION



Kubernetes, Slurm, serving schedulers, autoscalers, and fleet-management systems already own specialized control and execution decisions. Aurelius adds a pre-execution planning layer above them.

Before a decision is committed, Aurelius forks the current fleet state into competing simulated trajectories. Each trajectory applies a different coupled policy across the controls implemented for that workload, including batching, routing, placement, capacity, prewarming, admission, migration, precision, speculative decoding, and GPU clock policy.

The fleet state is then advanced through the planning horizon. Each trajectory changes queue progression, capacity use, placement and topology pressure, replica warm state, SLA outcomes, and modeled infrastructure cost.

Trajectories that violate configured SLA, capacity, power, residency, placement, or policy constraints are rejected before ranking. The highest-scoring feasible policy is returned to the existing control plane, which remains authoritative and executes through its normal interfaces.

8.24×

mean SLA-safe goodput per modeled infrastructure dollar versus a constructed production-informed replay baseline

84.5%

1,549,249

3 of 3

fewer GPU-hours under identical input request sets

requests across three selected uncapped high-load windows

windows with lower SLA violation rates

Simulated, load-dependent evidence from selected public-trace windows. Not a production result and not an estimate of performance against a particular fleet or internal scheduler.

The optimizer cannot choose a future it never represents.

The uncapped ablation reran the exact harness, windows, and load behind the 8.24× headline and varied one thing: the search strategy available to the planner. The production anchors reproduced the published values exactly, in all three markets.

FIG 02 SEARCH-STRATEGY ABLATION · UNCAPPED REPLAY			SINGLE VARIABLE · THE SEARCH	
SEARCH SPACE	COMPLETION	RELATIVE GP/\$	DELTA	MECHANISM
CONSTRUCTED BASELINE	COMPLETES	1.00×		REACTIVE REFERENCE
GPU CLOCK ONLY	TIMES OUT*	0.69×*	-31.3%*	SINGLE SURFACE
FIXED 24-POLICY GRID	COMPLETES	3.62×	+261.9%	JOINT POSTURE
REGIME-SPECIFIC CANDIDATE SET	COMPLETES	3.62×	+261.9%	GENERATED, NO SEARCH
EXHAUSTIVE SEARCH OF FIXED GRID	COMPLETES	4.84×	+383.6%	FULL GRID ENUMERATION
BOUNDED COUPLED SEARCH	COMPLETES	4.85×	+385.4%	BOUNDED BEAM
HIERARCHICAL COUPLED-POLICY SEARCH	COMPLETES	8.29×	+728.6%	OFF-GRID COUPLING
SLA VIOLATION RATE 0.038 TO 0.046 AT THE BASELINE, 0.0073 TO 0.011 FOR THE GRID ARMS, 0.0022 TO 0.0026 FOR THE HIERARCHICAL SEARCH · LOWEST OF ANY ARM. * TIMES OUT AT THE STANDARD HARNESS BUDGET; AT AN EXTENDED BUDGET IT COMPLETES BELOW THE BASELINE.				
IDENTICAL INPUTS, WINDOWS, AND LOAD IN EVERY ARM · 1,549,249 REQUESTS			RERUN · 07.2026	

A clock-only, single-surface search cannot be scored within the harness budget: the policy it selects degrades the simulated fleet until the replay exceeds it, in all three markets. Given far more compute it does complete, and still lands below the baseline (about 0.7×) with roughly ten times its SLA violation rate.

Setting the posture jointly is worth roughly 3.6× on its own: a fixed 24-policy grid and the regime-specific candidate set, scored without coupled search, select the same policy at this load.

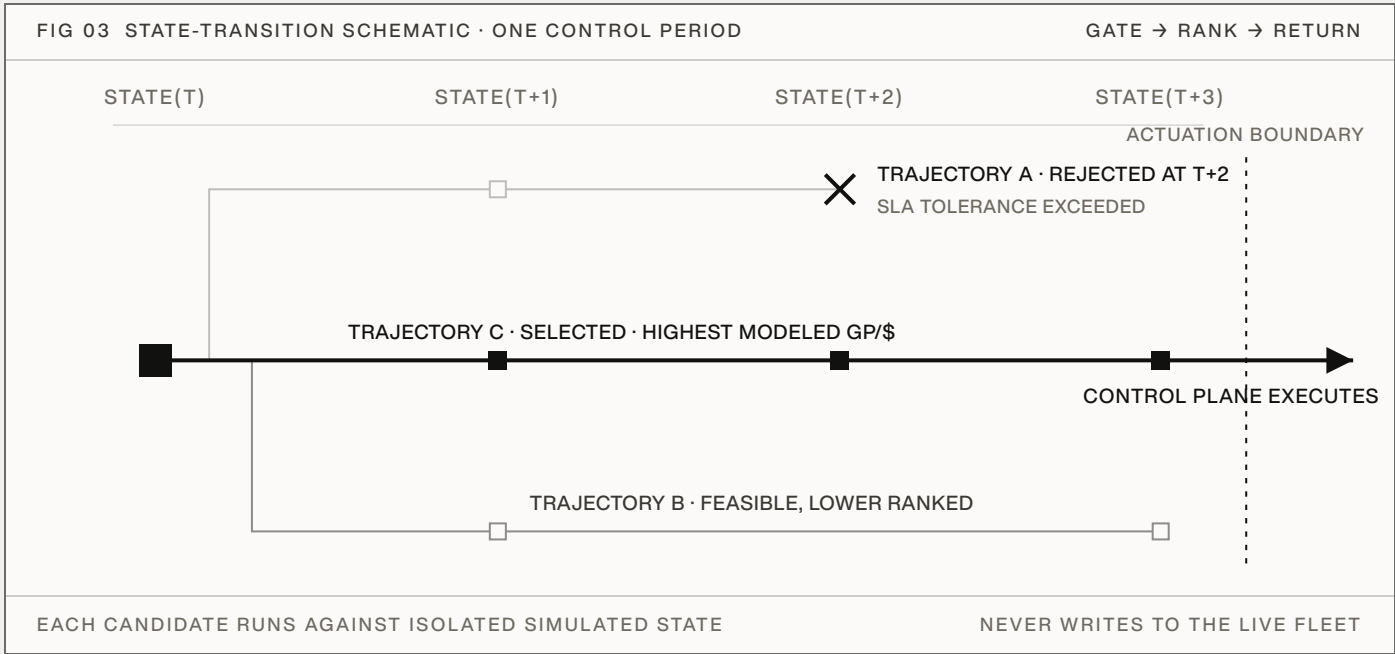
Searching the grid harder is saturated: exhaustive enumeration of the fixed grid reaches 4.84×, and the bounded coupled search matches it at 4.85× while evaluating far fewer candidates.

The hierarchical coupled-policy search reaches 8.29× with the lowest SLA violation rates of any arm. The near-doubling beyond every grid-based strategy is value in coupled futures no fixed grid represented. The forecast, simulator, objective, cost model, and constraint gates were unchanged in every arm.

The value was not hidden in a better score. It was hidden in a future the previous search never considered.

Uncapped search-strategy ablation, 07.2026: an independent rerun of the exact harness, windows, and load behind the headline benchmark. Production anchors reproduced published values exactly; the hierarchical arm reproduced the published headline within 1.2 percent per market (8.29× here against the published 8.24× mean). Ratios are unweighted means across the three markets. The frozen V1 audit separately measured zero search regret where exhaustive enumeration was tractable, at approximately 40 candidate evaluations per decision. Neither validates the simulator or establishes production savings.

One fleet. Many futures. Simulated before execution.



Live fleet state: queue, capacity, placement, topology, replica state, and configured constraints, advanced one control period at a time.

- 01

Mirror the fleet.

A persistent state-transition model carries queue state, capacity, placement and locality, replica warm and cold state, migrations in flight, configured constraints, and the modeled cost basis across control periods.
- 02

Generate coupled policies.

A regime-specific generator proposes a bounded set of policies based on the workload, operating conditions, implemented controls, and configured constraints. Previously validated candidates are retained as controls.
- 03

Advance each trajectory.

Every candidate is simulated independently over the planning horizon. Forecasts use only information available at the decision timestamp. Future arrivals, future prices, and future outcomes are not supplied to the planner.
- 04

Reject infeasible trajectories and return the winner.

Hard constraints reject infeasible trajectories before economic ranking. The highest-scoring feasible policy is returned as a recommendation. When forecast confidence falls below the configured threshold, Aurelius returns the configured deterministic fallback policy.

Aurelius sits above Kubernetes, Slurm, serving schedulers, routing systems, autoscalers, and fleet-management systems. It evaluates how their decisions interact across a shared planning horizon. It does not replace them, write cluster state, or bypass operator-configured constraints.

The proposed historical replay requires scheduler and fleet metadata. It does not require customer identity, request payloads, model outputs, or proprietary weights.

A signal too large to ignore and too simulated to trust without operator data.

FIG 04 UNCAPPED REPLAY · SLA-SAFE GOODPUT PER DOLLAR VS THE CONSTRUCTED BASELINE THREE MARKET WINDOWS



1,549,249 total input requests · 84.5% fewer total GPU-hours · lower SLA violation rates in all three windows · both policies received the identical input request sets.

The comparison policy is a constructed production-informed replay baseline: a deterministic public-replay policy assembled from established serving and cluster-control mechanisms, including continuous batching, deadline-aware ordering and admission, prefix-cache-aware routing, topology-aware placement, and backlog-driven capacity control. It is stronger than the naive SLA-aware reference, but it is not an operator's internal scheduler and does not reproduce a particular fleet's adaptive logic, internal signals, or cost model.

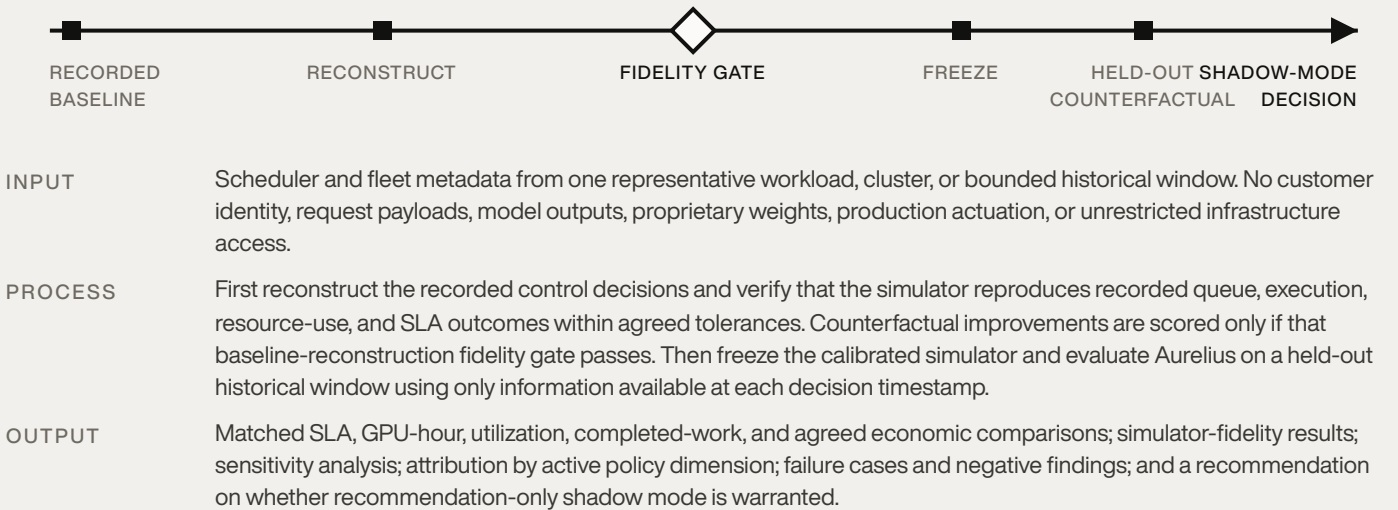
WHAT THIS ESTABLISHES

Under the selected uncapped high-load replay conditions, coupled predictive planning found feasible trajectories with materially better modeled economics and lower SLA violation rates than the constructed baseline.

WHAT THIS DOES NOT ESTABLISH

Expected performance against a particular fleet, internal scheduler, workload distribution, or operator cost model. The result is simulated and load-dependent. Its magnitude is materially affected by the benchmark's batching, precision, service-time, and modeled cost assumptions.

Test one bounded historical window.



REQUESTED NEXT STEP

A short scoping session with one engineering owner to select a bounded historical window, validate the minimum metadata schema, and agree on replay acceptance criteria.

If Aurelius cannot reproduce baseline behavior and improve the pre-registered primary metric on held-out telemetry, the evaluation stops. If it can, the next step is recommendation-only shadow mode.